MIL-T-85351A(AS)
25 January 1982
SUPERSEDING
MIL-T-85351(AS)
7 February 1980

MILITARY SPECIFICATION

TRANSPONDER SET AN/DPN-88

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture, testing, and acceptance of the AN/DPN-88 transponder set, referred to herein as the transponder.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-W-5088	Wiring, Aerospace Vehicle.
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for.
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment.
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-S-19500	Semiconductor Devices, General Specification for.
MIL-S-22710	Switches, Rotary (Printed Circuit), (Thumbwheel, Inline and Pushbutton), General Specification for.
MIL-M-38510	Microcircuits, General Specification for.

STANDARDS

FEDERAL

FED STD-595	Colors.

MILITARY

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701	List of Standard Semiconductors.
MIL-STD-704	Aircraft Electric Power Characteristics.

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MIL-STD-781	Reliability Design Qualifications and Production Acceptance Tests: Exponential Distribution.
MIL-STD-785	Reliability Program for System and Equipment Development and Production.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-883	Test Methods and Procedures for Microelectronics.
MIL-STD-965	Parts Control Program.
MIL-STD-1562	List of Standard Microcircuits.
MS 3114	Connectors, Receptacle, Electric, Series 1, Solder Type, Jam Nut Mounting, Bayonet Coupling.

DRAWINGS

MILITARY

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS3002	Transponder Set, AN/DPN-88

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

MILITARY

AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Reliability program. The contractor shall establish and maintain a reliability program in accordance with MIL-STD-785 that is planned, integrated and developed in conjunction with the transponder design, development, and production functions. An integrated reliability test and demonstration plan which shall include all reliability testing and demonstration to be performed during the program shall be prepared by the contractor for approval by the procuring agency at least 45 days in advance of testing (see 6.2.2).

3.2.1 Operational stability. The transponder shall operate as specified herein, continuously or intermittently, for a period of not less than 30 hours without the necessity of readjustment.

3.2.2 Operating life. The transponder shall have a total operating life of 200 hours with reasonable servicing and replacement of parts in accordance with MIL-STD-781.

3.2.3 Reliability in mean time between failure. The transponder shall have a mean time between failure (MTBF) of 50 hours when tested and accepted as specified herein (see 4.7).

3.3 Parts and materials. In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein (see 3.3.2).

b. Other parts and materials requirements shall be in accordance with MIL-E-5400 and MIL-STD-454.

c. Nonrepairable subassemblies shall be in accordance with AR-10 and MIL-E-5400.

d. When previously produced models of the transponder did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.3.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained in accordance with MIL-E-5400 and MIL-STD-965 (see 6.2.2).

3.3.2 Microelectronic device approval. Microelectronic devices shall be selected and approved in accordance with MIL-STD-965 and MIL-STD-883 as follows:

a. Devices that are listed in MIL-STD-1562 and are available from at least two qualified sources listed in a MIL-M-38510 QPL and purchased from a qualified source need not be submitted for approval nor tested for interchangeability.

b. The use of hybrid devices for which a second source is not available shall not relieve the contractor from obtaining and specifying the availability and interchangeability of no less than two sources for the chips and cases used therein.

3.3.3 Microelectronic and semiconductor selection. Microelectronic devices shall conform to the requirements of MIL-M-38510, Class B, as a minimum, with preference to MIL-STD-1562 types. Semiconductors shall conform to MIL-S-19500 with preference to MIL-STD-701 types. All microelectronic and semiconductor devices used shall be the full military temperature range (-55°C to +125°C).

3.3.4 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall be in accordance with MIL-STD-965.

3.3.5 Modules. The electronic portions of the transponder shall be functionally modularized in accordance with AR-10. Conformal coatings, encapsulants, embedments, or potting material used with modular assemblies containing integrated circuits and discrete parts shall be removable without damage to the assembly.

3.4 Major assemblies. The transponder shall contain the following major modules and assemblies:

a. RF assembly

b. Processor

c. Decoder

d. Encoder

e. Power supply

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f. Remote interface

g. Front panel control and connector assembly.

3.5 Associated equipment. The transponder shall operate with the following associated equipment:

a. AS-133/APX antenna assembly

b. AT-741/APX antenna assembly or equal.

3.6 Compatibility. The transponder shall be compatible for use with the ATC/IFF transponder test set VEGA Model 683L-1.

3.7 Cabling and connections.

3.7.1 Cables and connectors. The transponder shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.7.1.1 Interconnection cabling. The transponder shall operate as specified herein, using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the transponder.

3.8 Design and construction. The transponder shall be in accordance with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.8.1 Form factor. The dimensions of the transponder shall be as shown in Drawing 1356AS3002, which identifies the maximum volume available for the design of the transponder form factor. Where practical, the design shall permit module replacement without adjustments to the transponder.

3.8.2 Controls. The transponder shall be designed for local/remote reply code selection capability by mounting a four digit thumbwheel switch (see Drawing 1356AS3002), or equal, in accordance with MIL-S-22710, on the front panel to provide the following:

3.8.2.1 Mode selection. The left thumbwheel digit switch shall provide the means for selecting the interrogation Modes 1, 2, and 3.

3.8.2.2 Reply code selection. The two center thumbwheel digit switches shall be octal binary coded to provide the selection of any of the 32 possible reply code combinations from the front panel.

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3.8.2.3 Front panel remote selection. The right digit switch shall select whether the reply code will be from the front panel or from a remote source.

3.8.2.4 Electrical connections. Connections to external circuits shall be provided as shown in Table I.

TABLE I. Electrical connections.

Reference designation	Receptable type	Function
J1	MS3114H-12-10P	Power/remote
J2	TNC, 50 ohm, female	Antenna input
J3	BNC, 50 ohm, female	TACAN suppression

3.8.2.5 Pin connections. The signals shown in Table II shall appear at the designated pins of the power/remote connector J1.

TABLE II. J1 connector signals.

Pin number	Function
Pin A	+28 Vdc
Pin B	28 Vdc return
Pin C	Code select A4
Pin D	Code select A2
Pin E	Code select A1
Pin F	Code select B2
Pin G	Code select B1
Pin H	Isolated gnd return
Pin J	"X" pulse enable
Pin K	Gnd

3.8.2.6 Remote reply code programming. Six pins on connector J1 shall be required for remote programming, five for replay code selection, and one for return. An open collector interface driver or equal shall be required to program remotely each of the five reply code bits. A logical "zero" shall be represented by an open collector in saturation such that the collector emitter voltage will be less and or equal to +0.6 Vdc with +1.0 mA collector current. A logical "one" shall be represented by an open collector turned off such that the collector cutoff current will be less than 10.0 μ or with +5.0

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Vdc collector to emitter voltage. All interface power shall be supplied by an isolated supply within the unit. All signals shall be further isolated by high current-transfer-ratio optio-isolators.

3.9 Provisions for maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be in accordance with AR-10.

3.10 Performance. Unless otherwise specified herein, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified (see 4.6.2).

3.10.1 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than in accordance with MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, an Engineering Change Proposal (ECP) shall be prepared for review and approval by the procuring activity (see 6.2.2).

3.10.2 Warm-up time. The time required for the transponder to warm-up prior to operation shall be kept to a minimum and shall be not greater than two minutes under either standard or service conditions.

3.10.3 Operating power. The transponder shall perform as specified herein, when operated from the following power source having characteristics conforming to all applicable requirements of MIL-STD-704. The power required shall be not greater than the specified amounts as follows:

3.10.3.1 Direct current input voltage. The input voltage to the transponder shall be not less than +24 Vdc and not greater than +32 Vdc.

3.10.3.2 Direct current input power requirements. The input power shall be not greater than 40 watts over the input voltage range of +24 Vdc to +32 Vdc, when operating at maximum duty cycle.

3.10.3.3 Degraded performance. Degraded performance will be permitted for voltage transients not greater than 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the transponder when the transient has been removed.

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3.10.3.4 Input protection. The transponder shall not be damaged by external signals of any frequency or type of modulation of up to 10 volts peak amplitude and average power of -10 dBm or a +20 dBm peak power with a -30 dBm duty cycle applied directly to the antenna terminal.

3.10.4 Operation. The transponder shall receive coded interrogations and shall decode these interrogations and in response shall transmit proper pulse replies.

3.10.4.1 Interrogation characteristics. The transponder shall accept interrogation signals with characteristics as shown in Table III and Table IV.

TABLE III. Interrogation mode spacing.

Mode	Interrogation characteristic
1	2 pulses (P_1 and P_3) spaced 3 ±0.1 µsec
2	2 pulses (P_1 and P_3) spaced 5 ±0.2 µsec
3	2 pulses (P_1 and P_3) spaced 8 ±0.2 µsec

TABLE IV. Input pulse characteristics (all modes).

Characteristic	Value
Amplitude variation	$P_3 = P_1$ ±1 db
Pulse width	0.8 ±0.1 µsec
Rise time	0.05 to 0.1 µsec
Decay time	0.05 to 0.2 µsec

3.10.4.2 Interference characteristics. The transponder shall operate in the presence of the following types of interfering signals with minimum degradation:

3.10.4.2.1 Continuous wave and amplitude modulated continuous wave interference. The transponder shall operate in the presence of interfering signals consisting of continuous (CW) and amplitude-modulated continuous wave (AMCW) and shall meet the following requirements. Measurements shall be made to assure compliance throughout the required range of interrogating signal levels.

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a. If the interrogating signal to which the transponder shall reply is between 3 and 50 dB above the normal triggering level, an interfering signal 10 dB, below the interrogating signal shall not reduce the number of valid replies greater than 50 percent. At each level tested the interfering signal frequency shall be adjusted for maximum countdown of replies.

b. The AMCW signal shall be simulated using a signal generator modulated 30 percent by a 1000 Hertz (Hz) sine wave.

c. If the interrogating signal is between 20 and 50 dB above the normal triggering level, an interfering signal at a level 3 dB down from the interrogating signal shall not reduce the number of replies by greater than 50 percent, except when the interfering signal combines with the interrogating signal in such a way that the amplitude modulation of the resultant signal is inverted or eliminated.

d. In the presence of CW and AMCW interference as specified herein, the transponder shall not produce greater than 10 random or unsynchronized replies per second.

3.10.4.2.2 Pulse interference. The receiver shall operate in the presence of interfering signals consisting of randomly spaced pulses and regularly spaced pulses and shall meet the following requirements:

a. If the interrogating signal to which the transponder should reply is between 3 and 10 dB above the normal triggering level, an interfering signal 10 dB down from the interrogating signal shall not reduce the number of replies by greater than 50 percent.

b. If the interrogating signal is between 20 and 50 dB above the normal triggering level, an interfering signal 3 dB down from the interrogating signal shall not reduce the number of replies by greater than 50 percent.

c. If the interrogating signal consists of randomly spaced pulses only, and the interrogating signal is a Mode 1, 2 or 3 interrogation that is 10 dB above the interfering signal, the transponder shall reply to no less than 90 percent of the interrogations where the randomly spaced pulses are not recognized as normal or sidelobe interrogations.

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3.10.4.3 Echo rejection. The transponder shall contain echo rejection circuitry designed to permit normal operation in the presence of echoes of interrogation signals in space and be compatible with the requirements of 3.10.5.3.

3.10.4.3.1 Desensitization. Upon receipt of a pulse greater than 0.7 μsec in duration, the receiver shall be desensitized by an mount that is within at least 9 dB of the amplitude of the desensitizing pulse, but the amount of desensitization shall not at any time be greater than the amplitude of the desensitizing pulse with the exception of possible over-shoot during the first microsecond following the desensitizing pulse. Single pulses of duration less than 0.7 μsec are not required to cause the specified desensitization, and should not cause desensitization greater than those stated herein.

3.10.4.3.2 Recovery. The receiver shall remain desensitized for at least 2.5 μsec following reception of a desensitizing pulse of signal strength up to 50 dB above the minimum trigger level (MTL). The receiver sensitivity shall then recover linearly at a rate not greater than 4.5 dB per μsec to within 3 dB of MTL.

3.10.4.4 Minimum triggering level. Minimum triggering level is that level of signal required for 90 percent replies. The MTL of the transponder at the receiver antenna jack, when measured with nominal interrogation modes and at a frequency of 1030 ±0.2 MHz under standard conditions (see 4.3.1) shall be not less than -85 decibels referred to one volt (dBv) [-72 decibels referred to one milliwatt into 50 ohms (dBm) for all modes]. The total variation of MTL between Modes 1, 2 and 3 shall not exceed 1.0 dB when measured as above. Also the MTL of the transponder on Modes 1, 2 and 3 shall not change by more than ±1.0 dB when the interrogation pulsewidth is varied over the tolerance specified in 3.10.4.1 for each mode.

3.10.4.5 Triggering level drift. The triggering level, when set at -85 dBv (-72 dBm) under standard conditions (see 4.3.1) shall not vary by more than ±3 dB over all combinations of service conditions.

3.10.4.6 Random triggering rate. The transponder shall not transmit greater than 5 pulse trains per second averaged over 1 minute under the following conditions:

a. The receiver antenna receptacle is terminated in a 50 ohm resistive load.

b. No external signal is present.

c. Triggering level (sensitivity) control is set for an MTL of no less than -85 dBv (-72 dBm) as required in 3.10.4.4.

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3.10.4.7 Triggering level adjustment. The transponder MTL shall be internally adjustable between -82 dBv (-69 dBm) to -90 dBv (-77 dBm). It shall be adjusted to the level specified in 3.10.4.4 prior to delivery.

3.10.4.8 Reduced trigger level performance. The MTL as measured under conditions specified in 4.3.1 may be reduced no greater than 6.0 dB when the input voltage is reduced to 24 Vdc.

3.10.5 Overall performance requirements.

3.10.5.1 Automatic overload control. The automatic overload control (AOC) shall operate, by means of trigger level reduction, to perform the following functions:

3.10.5.1.1 Reply rate limiting. A sensitivity reduction type reply limit control circuit shall be incorporated in the transponder for the purpose of limiting the number of generated pulse trains in all modes without regard to the number of pulses in each reply. Sensitivity reduction greater than 3 dB shall not take effect until 90 percent of the selected reply rate is exceeded. Beyond the selected limit, sensitivity reduction shall be gradual to prevent weaker interrogations from denying transponder service to stronger interrogations as a result of over-interrogation. Sensitivity reduction shall prevent the reply rate from exceeding 125 percent of the selected value. The range of the reply limit control shall be adjustable to any value between 500 and 2000 replies per second. The reply rate limit shall be set at 1200 replies per second prior to delivery.

3.10.5.1.2 Sidelobe suppression rate limiter. This control shall limit the rate of sidelobe decoding on all modes by gradual reduction of triggering level. At a sidelobe suppression rate up to 5000 pulses per second there shall be no effect on the triggering level. With an interrogation signal 50 dB above MTL, maximum sidelobe suppression rate shall not be greater than 10,000 pulses per second. The sidelobe suppression rate limiter shall be activated only when either Mode 1, 2 or 3 is enabled.

3.10.5.1.3 Automatic overload control stability. The absolute limits stated in 3.10.5.1.2 shall apply under all service conditions. The rates established by adjustment in 3.10.5.1.1 under conditions specified in 4.3.1 shall hold within ±20 percent under all service conditions.

3.10.5.1.4 Reply rate limiting action and recovery time. If the interrogation rate is suddenly increased from 1000 to 2500 pulse pairs per second when the threshold established by the control described in 3.10.5.1.1 has been adjusted to 2000 replies per second, the sensitivity of the receiver shall be reduced by 20 dB, for interrogation pairs 20 dB above MTL, within 0.2 second. Upon decreasing the interrogation rate to 1000 pulse pairs per second the receiver shall return to within 3 dB of normal sensitivity within 0.025 seconds.

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3.10.5.2 Suppression characteristics.

3.10.5.2.1 Suppression during transmission of replies in modes 1, 2 and 3. Decoding of Modes 1, 2 and 3 shall be suppressed during transmission of replies in Modes 1, 2 and 3. Suppression shall continue for no less than 75 μsec after the second transmitted framing pulse, but no greater than 100 μsec.

3.10.5.2.2 Suppression input. Decoding of all modes shall be suppressed upon reception of a suppression pulse having the characteristics listed in Table V.

TABLE V. Suppression inputs pulse characteristics.

Characteristics	Value
Polarity	Positive
Amplitude	15 to 70V
Duration	1.5 to 250 μsec
Input impedance	2200 ±10% ohms resistance and 50 picofarad or less capacitance
Rise time	10 V/μsec or greater
Decay time	10 V/μsec or greater
Noise	No suppression shall result from any input of less than 5.0 volts peak amplitude.

3.10.5.2.2.1 Suppression operation. Upon application to the transponder of a suppression pulse having the above characteristics, decoding all modes shall be suppressed for an interval starting not greater than 1.0 μsec after the suppression pulse has reached an amplitude of 10 volts and shall continue for no less than the duration of the suppression pulse. After the end of the suppression pulse the receiver shall regain normal sensitivity, within 3 dB, no greater than 15 μsec after the suppression pulse has fallen to 5.0 volts. The transponder shall be insensitive to proper RF interrogations on all modes, with signal levels of -20 dBv, for the duration of the suppression pulse.

3.10.5.2.3 External suppression output. The transponder shall generate a suppression pulse during the time of transmission for each Modes 1, 2 and 3 output train, which is used to suppress other equipments. The suppression pulse shall have the characteristics specified in Table VI.

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TABLE VI. Suppression output pulse characteristics.

Characteristics	Value
Amplitude	18 to 30 V
Polarity	Positive
Synchronization	The pulse shall reach no less than 18 V by the time an RF output pulse has reached 10 percent of its detected amplitude.
Duration	The suppression pulse shall remain at 18 V or greater until the trailing edge of the last transmitted pulse has decayed to the 10 percent level. Suppression output shall also be less than 5.0 V, 5.0 µsec after the last transmitted pulse of the reply train has fallen to 10 percent level.
Rise time	20 V or more per µsec
Decay time	10 V or more per µsec
System impedance	The load impedance presented to the suppressor output shall be not greater than 2200 ohms nor less than 300 ohms shunted by not greater than 1850 pf capacitance.
Spurious output	Combined steady-state and transient voltages shall be not greater than 1.0 V either polarity, with the system loads described above.

3.10.5.3 Interrogation sidelobe suppression (ISLS).

3.10.5.3.1 ISLS control pulse characteristics. The ISLS control pulse will have characteristics as listed in Table VII.

TABLE VII. ISLS control pulse characteristics.

Characteristics	Value
Pulsewidth	0.8 ±0.1 µsec for Modes 1, 2 and 3 interrogations
Pulse position	2 ±0.15 µsec after the first pulse of a Mode 1, 2 and 3 interrogation.
Rise time	Between 0.05 and 0.1 µsec
Decay time	Between 0.05 and 0.2 µsec
Frequency	Within ±0.1 MHz of the frequency of the interrogation pulses (1030 MHz nominal).

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3.10.5.3.2 Conditions for ISLS. The decoding of Modes 1, 2 or 3 shall be suppressed not less than 99 percent efficiency for interrogations 3 dB or greater above MTL when the received amplitude of the ISLS control pulse is equal to or greater than the received amplitude of the first interrogation pulse and spaced 2 ±0.15 μsec from it. The suppression of the decoder shall be for a period of 35 ±10 μsec and shall be capable of being reinitiated for the full duration within 2 μsec after the end of the suppression period.

3.10.5.3.3 ISLS conditions for non-suppression. The suppression of the decoder shall not take place when the received amplitude of the first interrogation pulse is 9 dB or more in excess of the received amplitude of the ISLS control pulse, or when the spacing of the first interrogation pulse and the ISLS pulse is greater than 2.6 μsec.

3.10.5.3.4 ISLS gray region. If the level of the interrogation pulse is greater than the level of the suppression pulse but greater by less than 9 dB, suppression may or may not take place.

3.10.5.3.5 ISLS dynamic range. The ISLS suppression and non-suppression characteristics shall apply over a received signal amplitude range from 3 dB above MTL to greater than 50 dB above the level.

3.10.5.4 Pulse width discrimination. Signals of received amplitude between minimum triggering level and 6 dB above this level, and of a duration less than 0.3 μsec on Modes 1, 2 and 3 at the antenna input, shall not cause the transponder to initiate greater than 10 percent reply or suppression action. With the exception of pulses with amplitude variations approximating a normal interrogation or suppression pulse pair condition, any pulse or pair of pulses each of a duration greater than 1.5 μsec shall not cause the transponder to initiate reply or suppression action.

3.10.5.5 Transponder delay. The transponder delay between the second pulse of a Mode 1, 2 or 3 interrogation received at the antenna terminal and the first reply pulse delivered at the antenna terminal shall be between 2.5 and 3.5 μsec. The variation in delay between Modes 1, 2 and 3 shall be not greater than 0.2 μsec for nominally spaced interrogations.

3.10.5.6 Range jitter. On any one of Modes 1, 2 or 3 the total jitter of the reply pulse code group, with respect to P_3, shall be not greater than 0.1 μsec for receiver input levels between 3 and 50 dB above minimum triggering level.

3.10.5.7 Reply characteristics. The transmitted replies as measured at the antenna terminal shall have the characteristics shown in Table VIII.

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TABLE VIII. Reply pulse characteristics.

Characteristics	Value
Duration	0.45 ±0.1 µsec
Rise time	0.05 to 0.1 µsec
Decay time	0.05 to 0.2 µsec
Amplitude jitter	5 percent or less

3.10.5.7.1 Mode 1, 2 and 3 replies. For each pulse appearing at the decoder output, a pulse train shall be transmitted containing from none to a maximum of five information pulses plus two framing pulses which are spaced 20.3 +0.05 µsec. The information pulse spacing shall be in multiples of 2.9 µsec from the initial framing pulse with a maximum allowable deviation from this spacing of ±0.05 µsec. The position where a sixth pulse would appear (17.4 ±0.05 µsec from the initial framing pulse) shall not be used. From the specified five information pulses, a total of 32 different codes in each mode shall be available.

3.10.5.8 Code number assignments. Every configuration of Mode 1, 2 and 3 information pulses shall be assigned a code number.

3.10.5.8.1 Codes. Code numbers 00 through 73 shall designate the Mode 1, 2 and 3 codes. The applicable 5 information pulses shall be divided into 2 groups (A and B) with A containing the first 3 pulses and B containing the last 2 pulses. The information pulses in group A shall be numbered in order serially (1, 2 and 4) and the 2 pulses in group B shall be numbered in order (1 and 2). The pulse position measured from the initial framing pulse is shown in Table IX. The appropriate code number shall be obtained by adding the pulse numbers (1, 2, and 4) of the pulses present within each group and combining the sums in order (A, B) to form the serial code numbers.

TABLE IX. Modes 1, 2 and 3 pulse positions.

Pulse	Position from first framing pulse in µsec
A1	2.9 ±0.05
A2	5.8 ±0.05
A4	8.7 ±0.05
B1	11.6 ±0.05
B2	14.5 ±0.05

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3.10.5.8.2 X-pulse codes. A pulse spaced 10.15 ±0.05 μsec from the initial framing pulse shall be designated as the "X" pulse. The pulse position is normally unused. Provisions shall be made such that, when a single external control lead is grounded, all replies in Modes 1, 2 and 3 will include this pulse in addition to the normal framing and information pulses.

3.10.6 Receiver performance.

3.10.6.1 Receiver function. The receiver portion of the transponder shall receive RF energy from the antenna, amplify and decode it and initiate the proper transmitted pulses.

3.10.6.2 Receiver frequency. The receiver shall operate at a fixed frequency of 1030 MHz.

3.10.6.3 Receiver frequency stability. Under all combinations of service conditions specified herein, the frequency drift from that measured under standard conditions specified in 4.3.1 shall be not greater than ±1.5 MHz.

3.10.6.4 Receiver dynamic range. All performance requirements shall be met with interrogation signals from minimum triggering level to no less than 50 dB above that level unless otherwise specified herein.

3.10.6.5 Receiver bandwidth. Overall bandwidth shall be as specified in Table X. The bandpass shall be essentially gaussian and shall accommodate interrogation signal frequencies of 1030 ±2.5 MHz with no greater than 3 dB variation in sensitivity.

TABLE X. Overall receiver bandwidth.

Gain	Frequency
-6 dB points	> 7.0 MHz
-60 dB points	< ±25.0 MHz

3.10.6.6 Receiver RF rejection. The CW and pulse responses shall be 60 dB less than that measured at 1030 MHz for all frequencies less than 1005 MHz and greater than 1055 MHz.

3.10.7 Transmitter performance.

3.10.7.1 Transmitter function. The transmitter portion of the transponder shall generate RF power and transmit it to the load specified herein.

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3.10.7.2 Transmitter frequency. The transmitter shall operate at a fixed frequency of 1090 MHz.

3.10.7.3 Transmitter frequency stability. When the transmitter has been adjusted under the standard conditions specified in 4.3.1 and then submitted to any combination of service conditions, the transmitter frequency shall remain at 1090 ±3.0 MHz.

3.10.7.4 Transmitter power output. The transmitter shall deliver 27 ±3 dB above one watt peak power to the antenna terminal of the transponder under all combinations of conditions specified herein.

3.10.7.5 Transmitter power output variation within a pulse train. The peak power of any pulse in a group of pulses specified in 3.10.5.7 shall not differ from the power of any other pulse in that group by greater than 1.0 dB.

3.10.7.6 Transmitter duty cycle. The transmitter shall be capable of continuous operation at 1.0 percent duty cycle. For mean-time-between-failures (MTBF) and life-test considerations, operations shall be at 0.5 percent duty cycle for 50 minutes and 1.0 percent for 10 minutes of each hour. For first article acceptance testing, normal cycle shall be 0.5 percent and limit conditions shall be 0.1 percent and 1.0 percent duty cycles.

3.10.7.6.1 Duty cycle. The percent duty cycle shall be determined by the following equation:

$$DC = \frac{(PW) \times (RC) \times (NP)}{10^4}$$

where;
DC is percent duty cycle
PW is pulsewidth in microseconds
RC is reply count in number of counts
NP is number of pulses

3.10.8 Decoder performance. The decoder shall meet the following requirements:

3.10.8.1 Decoder operation. The decoder shall distinguish and cause replies to the three modes of interrogation specified in 3.10.4.1. In addition, interrogations made up of pulses of 0.6 to 1.2 μsec width with rise times of 0.2 μsec and decay times of 0.3 μsec, shall be decoded and replies generated with no degradation in performance, except that the decoder tolerance may be proportional to pulsewidth. This circuitry shall be such that the reply transmitted by the transponder shall be the same regardless of the mode of interrogation.

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3.10.8.2 Decoder tolerance, mode 1. Interrogation pulses spaced 2.8 to 3.2 µsec shall cause the equipment to generate Mode 1 replies. Pulses spaced less than 2.5 or greater than 3.8 µsec shall not cause such replies.

3.10.8.3 Decoder tolerance, mode 2. Interrogation pulses spaced 4.8 to 5.2 µsec shall cause the transponder to generate Mode 2 replies. Pulses spaced less than 4.2 or greater than 5.8 µsec shall not cause such replies.

3.10.8.4 Decoder tolerance, mode 3. Interrogation pulses spaced 7.8 to 8.2 µsec shall cause the transponder to generate Mode 3 replies. Pulses spaced less than 7.2 or greater than 8.8 µsec shall not cause such replies.

3.10.8.5 Mode selection. The three modes of interrogation shall be selectable by means of an appropriate switch located on the front of the transponder.

3.10.9 Power supply performance. The power supply shall meet the following requirements:

3.10.9.1 Power supply design. An integral power supply shall be provided to accept input power as specified in 3.10.3 and provide proper conversion to the voltages and currents required by the remainder of the transponder. This supply shall be of solid state, modular, quick-replaceable design and shall incorporate to the maximum extent possible nonrepairable, easy-replaceable subassemblies to facilitate maintenance by the substitution methods.

3.10.9.2 Power supply protection. The power supply shall be self-protecting against faults which place an overload or short circuit on any or all of the power supply outputs. The power supply need not operate during the presence of such faults, but the use of fuses for this protection will not be allowed, and operation shall return to normal once the fault has cleared. A reversal in line input voltage polarity, applied continuously, or an instantaneous drop in the line voltage to zero potential shall not damage the transponder.

3.10.10 Antenna mismatch. Anticipated service conditions of impedance mismatch shall include all values of load impedance that present a voltage standing wave ratio (VSWR) up to and including 1.5:1 of any phase relationship at the antenna terminal of the receiver-transmitter unit.

3.11 Environmental requirements.

3.11.1 Service conditions. The transponder shall operate as specified herein under any of the environmental service conditions or combination of these.

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3.11.1.1 Temperature/altitude. The transponder shall operate satisfactorily when subjected to the temperature/altitude requirements of MIL-E-5400, Table I, Class 2, except the altitude shall be to 100,000 feet as shown in Figure 1 herein.

3.11.1.2 Humidity. The transponder shall operate satisfactorily when subjected to the humidity tests.

3.11.1.3 Leakage (immersion). The transponder shall operate satisfactorily when subjected to the leakage tests.

3.11.1.4 Vibration. The transponder shall operate satisfactorily when subjected to the random vibration tests.

3.11.1.5 Acceleration. The transponder shall operate satisfactorily when subjected to accelerations of 20 Gs for a duration of one minute.

3.11.1.6 Shock. The transponder shall operate satisfactorily when subjected to shock conditions with peak amplitudes of 20 Gs for a duration of 11 milliseconds.

3.12 Electromagnetic interference. When specified in the contract or purchase order (see 6.2.1), the transponder shall meet the electromagnetic interference requirements of MIL-STD-461 for Class A1 equipment, Methods CE01, CE02, CE03, CE04, CE06, CS01, CS02, CS03, CS04, RE02, and RS03.

3.13 Weight. The total weight of the transponder excluding cables, shall be a minimum consistent with good design and shall be not greater than five pounds.

3.14 Finish and color. The transponder shall be semi-gloss blue, color number 25053 in accordance with FED-STD-595. Lettering shall be on labels with natural metal characters and lusterless black, color number 37038 background in accordance with FED-STD-595.

3.15 Nomenclature, nameplates and identification marking. A nameplate shall be used to mark the transponder. The marking information shall be in accordance with MIL-STD-130 and Drawing 1356AS3002. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.16 Workmanship. The transponder shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9, and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

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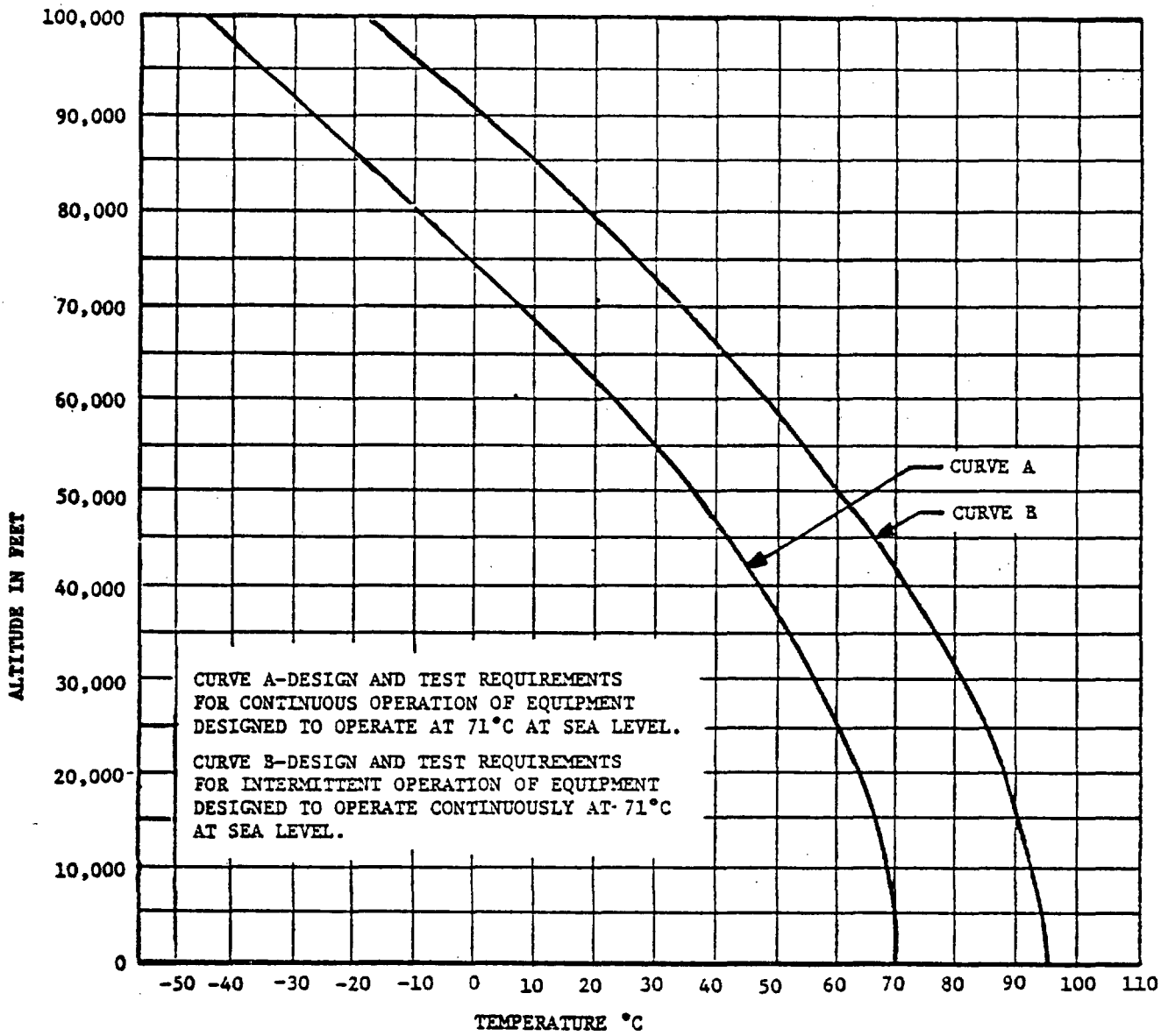


FIGURE 1. Operational requirements for class 2 airborne electronic equipment (temperature vs altitude).

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all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article test (see 4.4).

b. Initial production test (see 4.5).

c. Acceptance tests (see 4.6).

d. Reliability test (see 4.7).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the following test conditions.

4.3.1 Standard conditions. The following conditions shall be used to establish normal performance requirements under standard conditions and making laboratory bench tests:

a.	Temperature:	Room ambient (25°C ±10°C)
b.	Altitude:	Normal ground
c.	Vibration:	None
d.	Humidity:	Room ambient up to 90 percent relative humidity
e.	Input power voltage:	27.5 ±0.5 Vdc
f.	Interrogation rate:	500 interrogations per second, when not otherwise specified herein.

4.4 First article test. When a first article is required (see 6.2.1), first article tests shall be conducted by the contractor on two transponders. The transponders shall be representative of the production transponders to be supplied under the contract. Tests shall be accomplished under the approved test procedures of 4.8. The Government representative and the procuring activity shall be advised approximately 30 days in advance of testing so that a Government representative may be designated to witness tests when so desired.

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4.4.1 First article test data. The contractor shall prepare all data collected during the first article acceptance tests for review and approval by the procuring activity (see 6.2.2).

4.4.2 Scope of tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the transponder meets all the requirements of this specification, other applicable specifications and the contract. The first article tests shall include the quality assurance tests specified in Table XI, to provide evidence of compliance with the requirements of Section 3, electrical bench tests (see 4.4.2.1), environmental tests (see 4.4.2.2), and electromagnetic interference tests (see 4.4.3).

4.4.2.1 Electrical bench tests. Electrical bench tests shall be conducted to demonstrate compliance with the performance requirements specified herein. The bench tests shall be conducted under standard conditions (see 4.3.1).

4.4.2.2 Environmental tests. Environmental tests shall be conducted to demonstrate compliance with the environmental requirements specified herein. Tests shall include, but not necessarily limited to, the following environmental test methods in accordance with MIL-STD-810. Prior to preceding with any of the environmental tests the guidelines specified in MIL-STD-810, Section 3.2, shall be adhered to. Environmental testing shall be conducted on the first article as follows:

TABLE XI. Quality assurance tests.

Test	Requirement	Requirement paragraph	Method paragraph
Examination of product	Parts and material	3.3	4.6.1
	Design and construction	3.8	4.6.1
	Electrical connections	3.8.2.4	4.6.1
Performance	Input electrical power	3.10.3.2	4.6.2
	Interrogation characteristics	3.10.4.1	4.6.2
	Interference characteristics	3.10.4.2	4.6.2
	Minimum triggering lever	3.10.4.4	4.6.2
	Suppression characteristics	3.10.5.2	4.6.2
	Interrogation sidelobe suppression (ISLS)	3.10.5.3	4.6.2
	Pulsewidth discrimination	3.10.5.4	4.6.2
	Reply characteristics	3.10.5.7	4.6.2
	Receiver performance	3.10.6	4.6.2
	Transmitter performance	3.10.7	4.6.2
	Decoder performance	3.10.8	4.6.2
	Power supply performance	3.10.9	4.6.2
Environmental	Service conditions	3.10.11	4.4.2.2

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4.4.2.2.1 Temperature/altitude. A temperature/altitude test shall be conducted in accordance with MIL-STD-810, Methods 504.1, Procedure I for category 6 equipment except as follows:

a. The altitude shall be to 100,000 feet.

b. Use Figure 1 herein, in place of Figure 504.1-2 of MIL-STD-810.

4.4.2.2.2 Humidity. A humidity test shall be conducted in accordance with MIL-STD-810, Method 507.1, Procedure I.

4.4.2.2.3 Leakage (immersion). A leakage test shall be conducted in accordance with MIL-STD-810, Method 512.1, Procedure I. The transponder shall be nonoperating.

4.4.2.2.4 Vibration. A random vibration test shall be conducted in accordance with MIL-STD-810, Method 514.2, Category (b.2), Procedure IA, Table 514.2 IIA, Figure 514.2-2A, without isolators.

4.4.2.2.5 Acceleration. An acceleration tests shall be conducted in accordance with MIL-STD-810, Method 513.2, Procedure II. The transponder shall be operating for one minute at 20 Gs accelerations in each of six directions.

4.4.2.2.6 Shock. A shock test shall be conducted in accordance with MIL-STD-810, Method 516.2, Procedure I. The transponder shall be operating.

4.4.3 Electromagnetic interference. An electromagnetic interference test shall be conducted in accordance with MIL-STD-462, Methods CE01, CE02, CE03, CE04, CE06, CS01, CS02, CS03, CS04, RE02, RS02, and RS03. Dilution of the requirements of 3.12 herein by the contractor, or tailoring of tests less than intended use or frequency of operation, shall be prepared in writing with justification for the change in test procedures for review and approval by the procuring activity (see 6.2.2).

4.4.4 First article approval. Approval of the first article shall be by the procuring activity upon satisfactory completion of all tests. No production transponders shall be delivered prior to the approval of the first article.

4.5 Initial production tests. Unless otherwise specified in the contract or purchase order (see 6.2.1), one transponder of the initial production lot shall be tested in accordance with 4.6.1 through 4.6.3 herein.

4.5.1 Initial production electromagnetic interference tests. When specified in the contract or purchase order (see 6.2.1), the contractor shall perform the tests of 4.4.3 herein.

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4.6 Acceptance tests. The following acceptance tests shall be performed on the transponder to determine compliance with the requirements of materials, workmanship, operation, and reliability as specified herein.

4.6.1 Examination of transponder. The contractor shall conduct an examination on each transponder in accordance with Table XI, to assure acceptability of the transponder.

4.6.2 Operational test. Each transponder shall be operated under standard conditions for no less than ten minutes. Following this warm-up period, the contractor shall conduct an operational tests in accordance with the tests specified in Table XI, to assure satisfactory transponder performance. The measured characteristic data shall be recorded on a data sheet for review and approval by the procuring activity for each transponder delivered (see 6.2.2).

4.6.3 Operating run-in test. Each transponder shall be operated under the conditions specified herein for a period of one hour without experiencing failure in the characteristics measured and recorded in 4.6.2 herein. The transponder shall be vibrated, without vibration isolators, for a period of ten minutes prior to the one hour test. The transponder shall be nonoperating during the vibration period. A record of failures and causes shall be maintained for review by the procuring activity. Transponders which fail to satisfy requirements shall be repaired and retested. The environmental conditions for the test shall be as follows:

a. Temperature: Room ambient.

b. Humidity: Room ambient.

c. Vibration: Any selected frequency within the range of 20 to 30 Hz, excluding resonant points, and an amplitude producing no less than 3 Gs normal to the mounting surface.

4.7 Reliability tests. The reliability test performed on the transponder for the initial production lot shall be conducted in the following manner:

a. The acceptance/rejection criteria shall be in accordance with MIL-STD-781, Test Plan IV C.

b. The transponder shall be operated in the following environmental tests for simulating typical mission profiles:

(1) The first phase of the cycle shall consist of a combined temperature-vibration environment:

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(a) The transponder shall be at ambient conditions and the temperature transitioned over 0.5 hour to 58°C where it will be maintained for three hours. The temperature will then be reduced to -40°C over a one hour period and shall be maintained at that temperature for three hours. The temperature will then be increased to ambient during the next 0.5 hour. Vibration of 2.2 Gs at any nonresonant frequency between 20 and 30 Hz, shall be induced during the entire cycle.

(2) The second phase shall consist of transponder operation at 50,000 feet simulated altitude for one hour. This phase may include time to reach altitude and return to ambient.

c. The first cycle and every third cycle from the first shall be conducted using 25 Vdc; the second cycle and every third cycle from the second shall use 28 Vdc; and the third cycle and every third cycle from this cycle shall use 31 Vdc.

d. During each cycle the following tests shall be performed and associated data recorded:

(1) External suppression output (see 3.10.5.2.3)

(2) ISLS (see 3.10.5.3.2)

(3) Coder performance, except 3 codes shall be randomly selected (see 3.10.5.8.1)

(4) Receiver frequency (see 3.10.6.2)

(5) Decoder performance, Mode 2 only (see 3.10.8.3)

(6) Transmitter performance (see 3.10.7.2, 3.10.7.3, and 3.10.7.4).

e. Detailed requirements and conditions preceding the reliability test are specified in MIL-STD-781, Paragraph 5. The transponder shall be tested until an accept or a reject decision is reached. Each transponder selected for the reliability test must have been subjected to and have successfully passed the acceptance tests of 4.6.1 through 4.6.3.

4.8 Testing procedures. Test procedures shall be prepared for review and approval by the procuring activity. The test procedures shall apply only to the complete transponder. These test procedures shall be prepared for review no less than 45 days in advance of testing. The format for the test procedures shall be in accordance with MIL-T-18303 (see 6.2.2).

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4.9 Test reports. A report on the results of all tests performed by the contractor on the transponder shall be prepared for reivew by the procuring activity in accordance with MIL-STD-831 (see 6.2.2).

5. PACKAGING

5.1 Level of packaging. The level of packaging shall be as specified in the contract or purchase order (see 6.2.1), and shall be in accordance with MIL-E-17555 and MIL-STD-794.

6. NOTES

6.1 Intended use. The transponder is intended to provide automatic radar identification in pilotless airborne vehicles to all suitably equipped, challenging aircraft, ships and ground forces operating within the operational range of the transponder.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirement for first article (see 3.1 and 4.4).

c. Electromagnetic interference requirements (see 3.12 and 4.5.1).

d. Responsibility for inspection (see 4.1).

e. Number of transponders for initial production tests if different than specified (see 4.5).

f. Level of packaging (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph No.	Data requirement title	Applicable DID No.	Option
3.2	Reliability program	DI-R-7079	
3.3.1	Nonstandard part approval requests/proposed attitions to an approved PPSL	DI-E-7029	
3.10.1	Engineering change proposal	DI-E-4527A	
3.15	Government nomenclature, nameplate approval, serial number request	UDI-P-22502B	
4.4	First article inspection procedures	DI-T-4901	
4.4.1	First article inspection report	DI-T-4902	
4.4.3	Electromagnetic interference test procedures	DI-R-7063	
4.4.3	Electromagnetic interference test reports	DI-R-7062	
4.6.2	Operational test report	DI-T-5329	
4.7.1	Reliability test reports	DI-R-7034	
4.8	Test procedures	UDI-T-21347	
4.9	Test reports	DI-T-2072	

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project 5895-N301)

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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